

Rhilo Sotto

COMPE470

1 November 2024

Assignment 3

Modules

Behavioral

```
module FIRfilterBehavioral #(parameter WL = 4, a0 = -3, a1 = 3, a2 = 4)
(input CLK, input signed [WL-1: 0] x,
output signed [2*WL-1:0] y);

reg signed [WL-1:0] R0, R1;

    always @(posedge CLK) begin
        R0 <= x;
        R1 <= R0;
    end

    assign y = (x * a0) + (R0 * a1) + (R1 * a2);

endmodule
```

Structural

```
module FIRfilterStructural #(parameter WL = 4, a0 = -3, a1 = 3, a2 = 4)
(input CLK, input signed [WL-1: 0] x,
output signed [2*WL-1:0] y);
```

```
wire [WL-1:0] r0, r1;
wire [2*WL-1:0] m0, m1, m2;
wire [2*WL-1:0] p0;
```

```
Register #(.WL(WL)) R0(.CLK(CLK), .x(x), .y(r0)),
            R1(.CLK(CLK), .x(r0), .y(r1));
```

```
SignedMultiplier #(.WL(WL), .coeff(a0)) SM0(.x(x), .y(m0));
SignedMultiplier #(.WL(WL), .coeff(a1)) SM1(.x(r0), .y(m1));
SignedMultiplier #(.WL(WL), .coeff(a2)) SM2(.x(r1), .y(m2));
```

```
SignedAdder #(.WL(2*WL)) SA0(.x(m0), .y(m1), .sum(p0)),
              SA1(.x(p0), .y(m2), .sum(y));
```

```
endmodule
```

Testbenches

```
`timescale 1ns / 1ns

module tb_FIRfilter;

localparam WORDLENGTH = 5;

reg CLK;
reg signed [WORDLENGTH-1:0] x;
wire signed [2*WORDLENGTH-1:0] y_b, y_s;

// clock signal
parameter ClockPeriod = 10;
initial CLK = 0;
always #(ClockPeriod / 2) CLK = ~CLK;

// instantiate DUTs with parameter
FIRfilterBehavioral #(.WL(WORDLENGTH)) FIR0 (.CLK(CLK), .x(x), .y(y_b));
FIRfilterStructural #(.WL(WORDLENGTH)) FIR1 (.CLK(CLK), .x(x), .y(y_s));

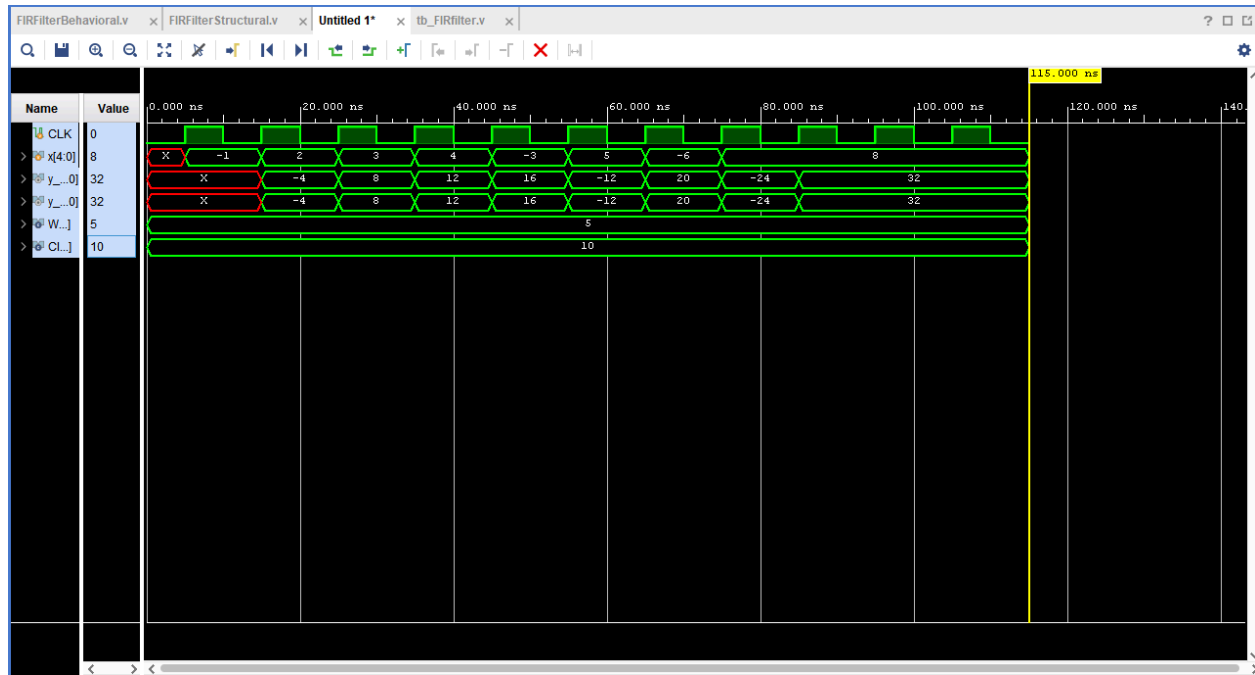
initial begin
    @(posedge CLK) x = -1;
    @(posedge CLK) x = 2;
    @(posedge CLK) x = 3;
    @(posedge CLK) x = 4;
    @(posedge CLK) x = -3;
    @(posedge CLK) x = 5;
    @(posedge CLK) x = -6;
    @(posedge CLK) x = 8;
    #(ClockPeriod * 4) $finish;

end

endmodule
```

Waveforms

Datapath



Corresponding to the modules and testbench, the output is as expected.